Midterm: MECHENG 4 AI 3

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1. Model Training

We trained a deep neural network model using Google Teachable Machine with the following parameters:

epochs = 8, batch size = 16, and learning rate = 0.01

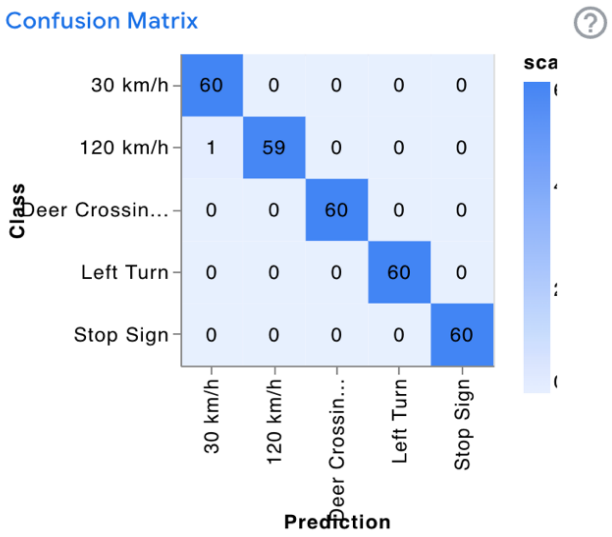
On the provided Stop Signs dataset, which consists of Traffic Signs images broken down into the 5 following classes:

- 30 km/h
- 120 km/h
- Deer Crossing
- Left Turn
- Stop Sign

The model is trained using the following hyperparameters:

epochs = 8, batch size = 16, and learning rate = 0.01

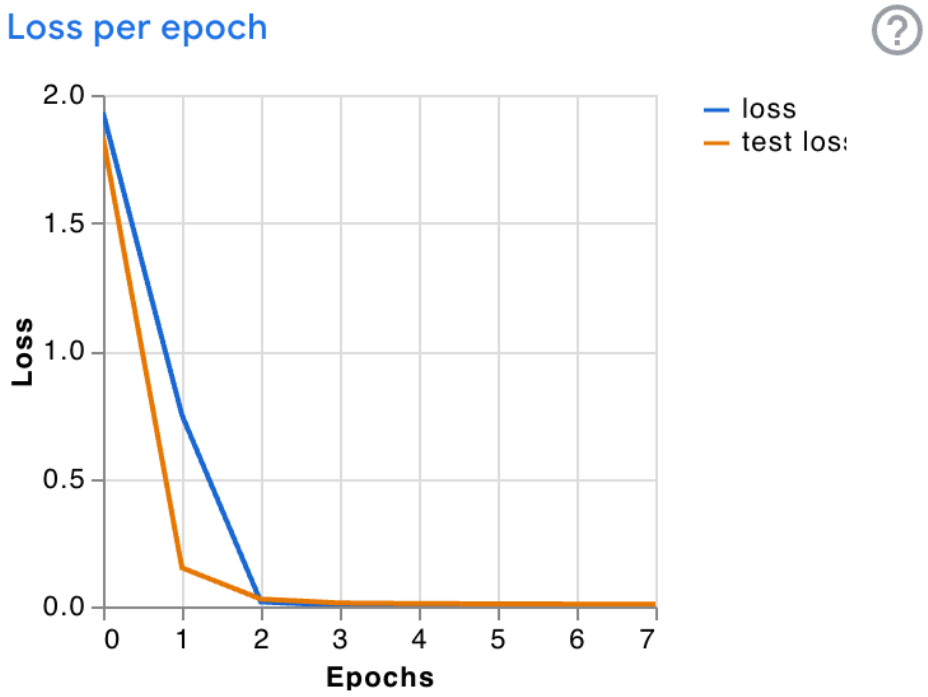
2. Evaluate Model Performance



Confusion matrix

The confusion matrix shows that the model performs almost perfectly across all classes. There is only one confusion made is on the class 120 Km/h with 30 Km/h. Which is a relevant error.

3. Loss Curve Analysis



Loss Curve

Both the training and testing loss curves decrease gradually and converge, which indicates stable learning. No overfitting nor underfitting is detected.

5. Hyperparameter Tuning

We tested the above combination of the different hyperparameters: (See Appendix for the figures)

1. Epoch = 8, batch size = 16, learning rate = 0.001
   The model achieved perfect classification with no errors across all classes. Both training and testing losses converge smoothly. Although perfect accuracy can indicate overfitting, the similar behavior of train and test loss suggests good generalization.

2. Epoch = 8, batch size = 32, learning rate = 0.001
   The results are identical to the first configuration. The model achieved 100 percent accuracy

on all classes. Training and testing losses decrease and converge together, indicating stable learning.

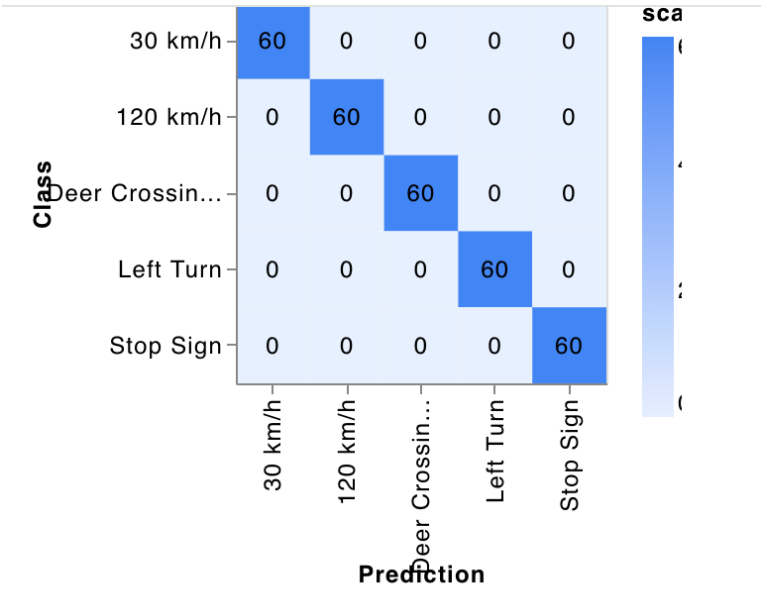
3. Epoch = 16, batch size = 32, learning rate = 0.01
   The model made one misclassification, where a Stop sign was predicted as 30 km/h. Overall performance remains extremely high. The loss converges for both datasets, but the test loss is slightly higher, indicating minor generalization gap.

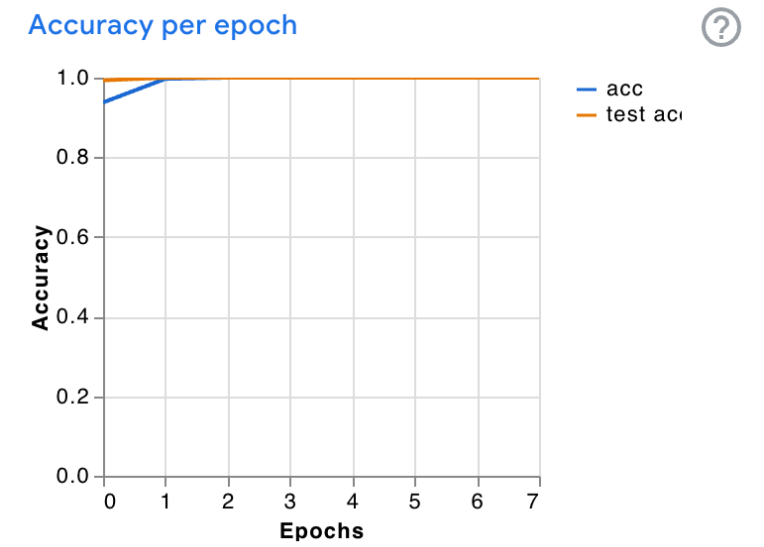
4. Epoch = 16, batch size = 128, learning rate = 0.001
   The model again reached perfect accuracy across all classes. The loss curves for training and testing converge consistently. No visible overfitting appears.

5. Epoch = 10, batch size = 128, learning rate = 0.001
   The model achieved perfect classification once more. Training and testing losses converge together, suggesting stable optimization and strong generalization performance.

Performance on Holdout Testing Dataset

By testing on the sample test image on each class, the model generalized very well and predict with a high confidence probability: around 98% everytime. There is no discrepancies between training and testing performance.





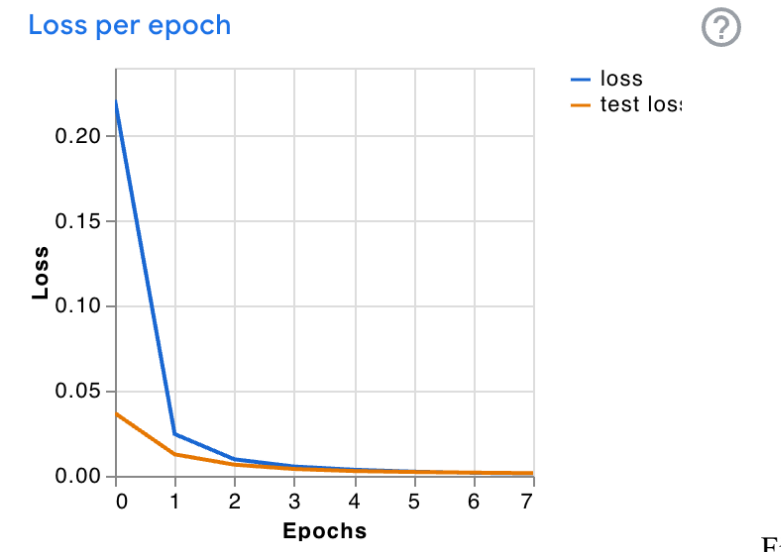
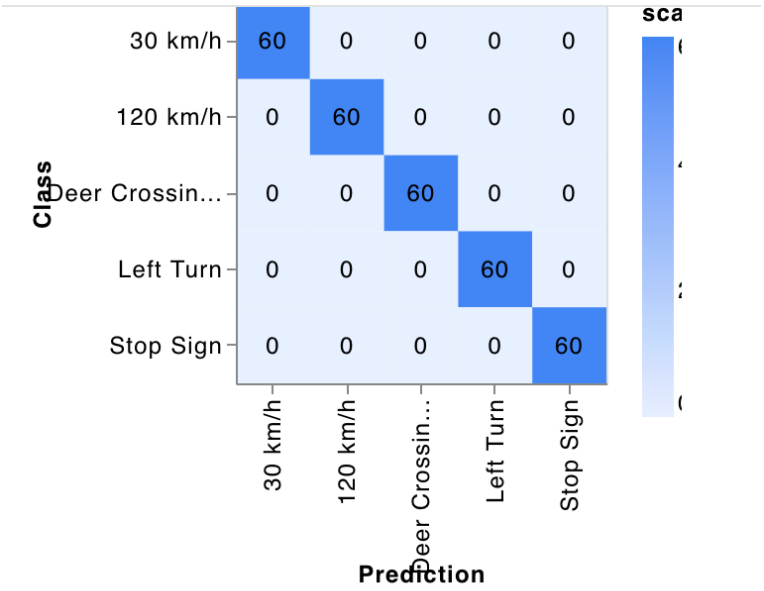
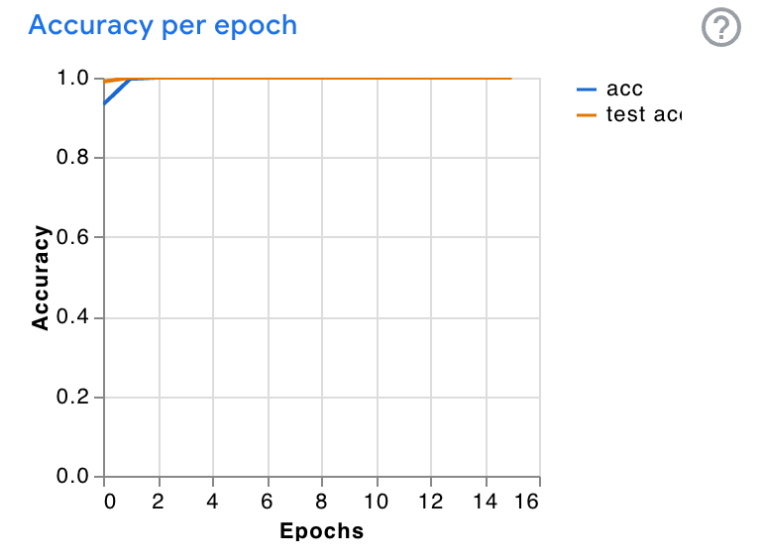


Figure 1 Combination 1





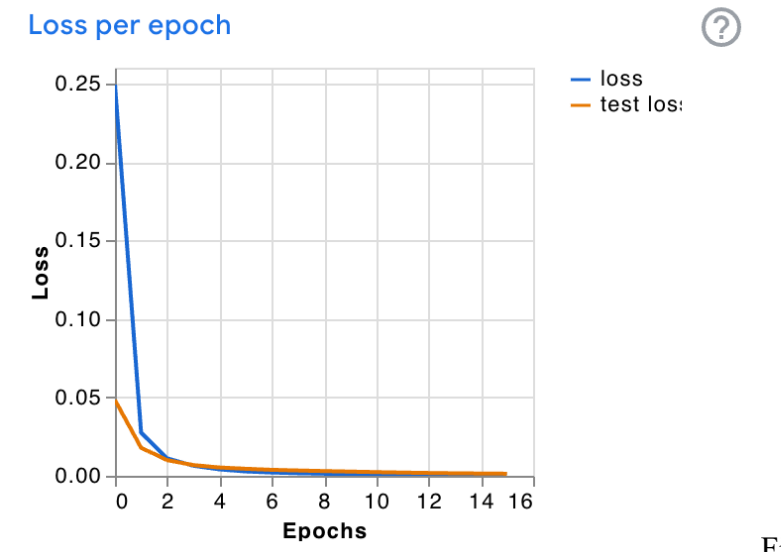
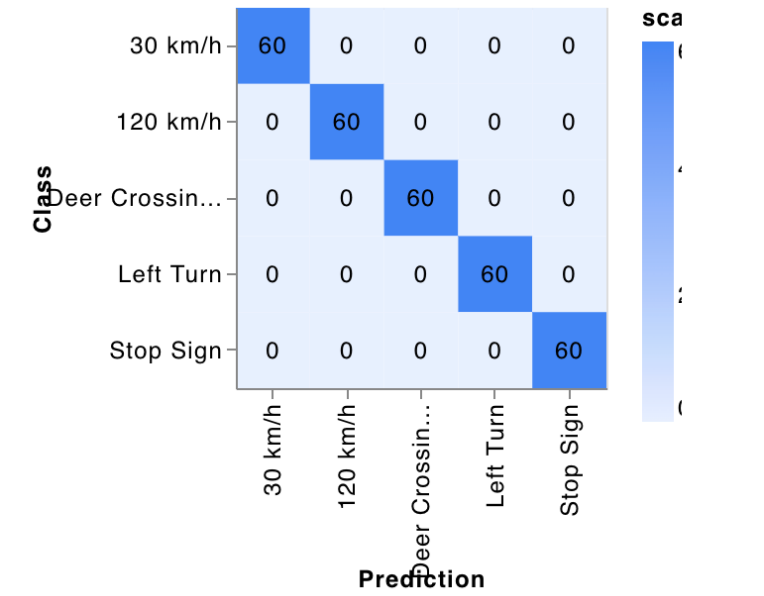
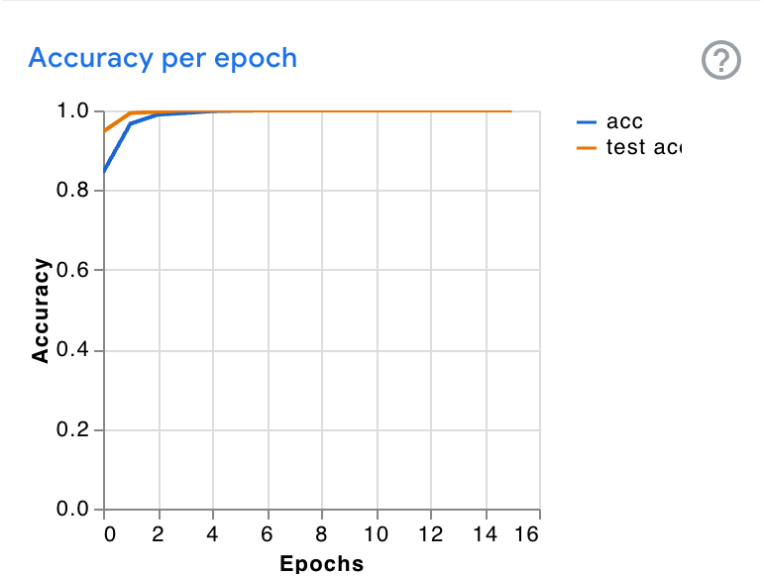


Figure 2 Combination 2





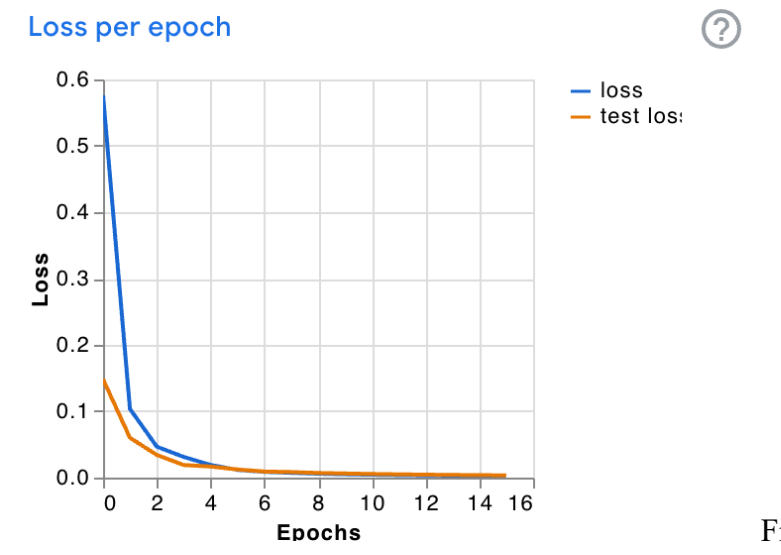
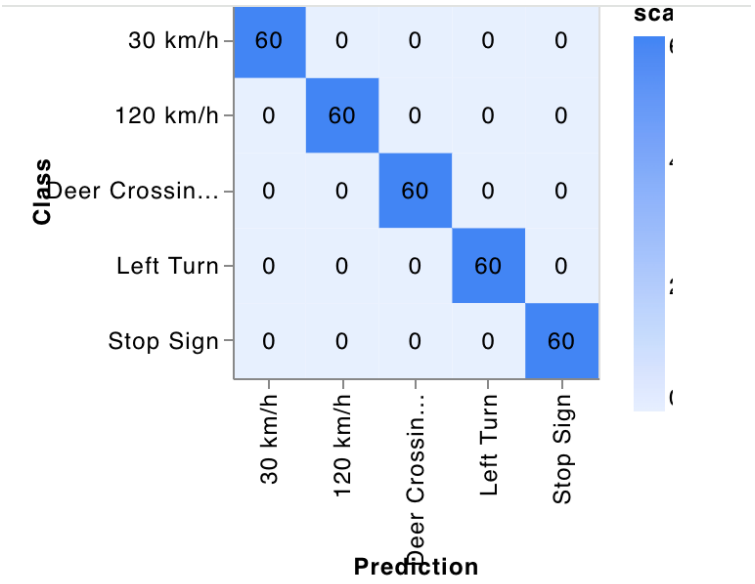
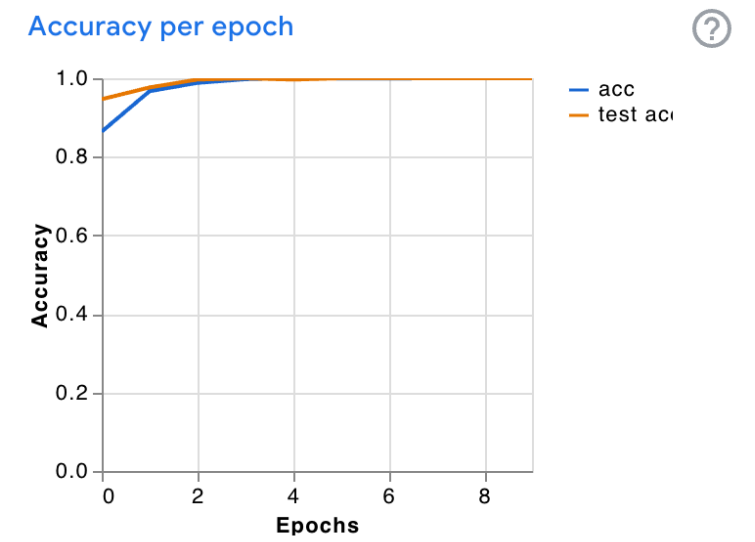


Figure 3 Combination 3





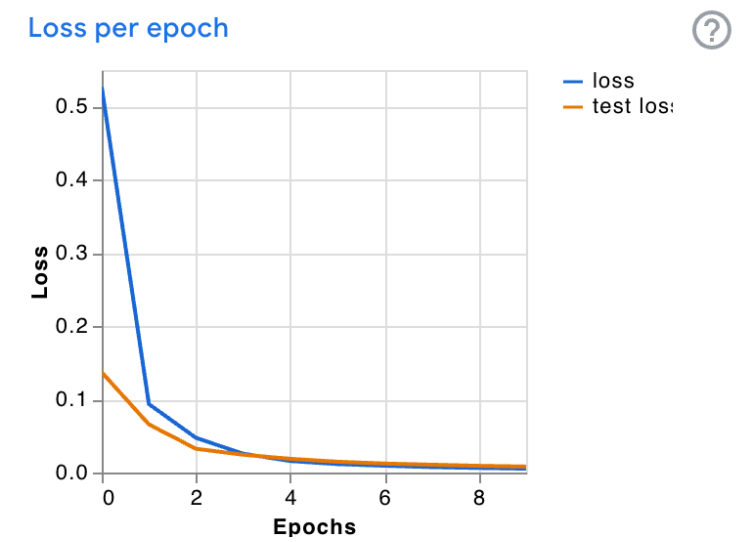


Figure 4 comination 4

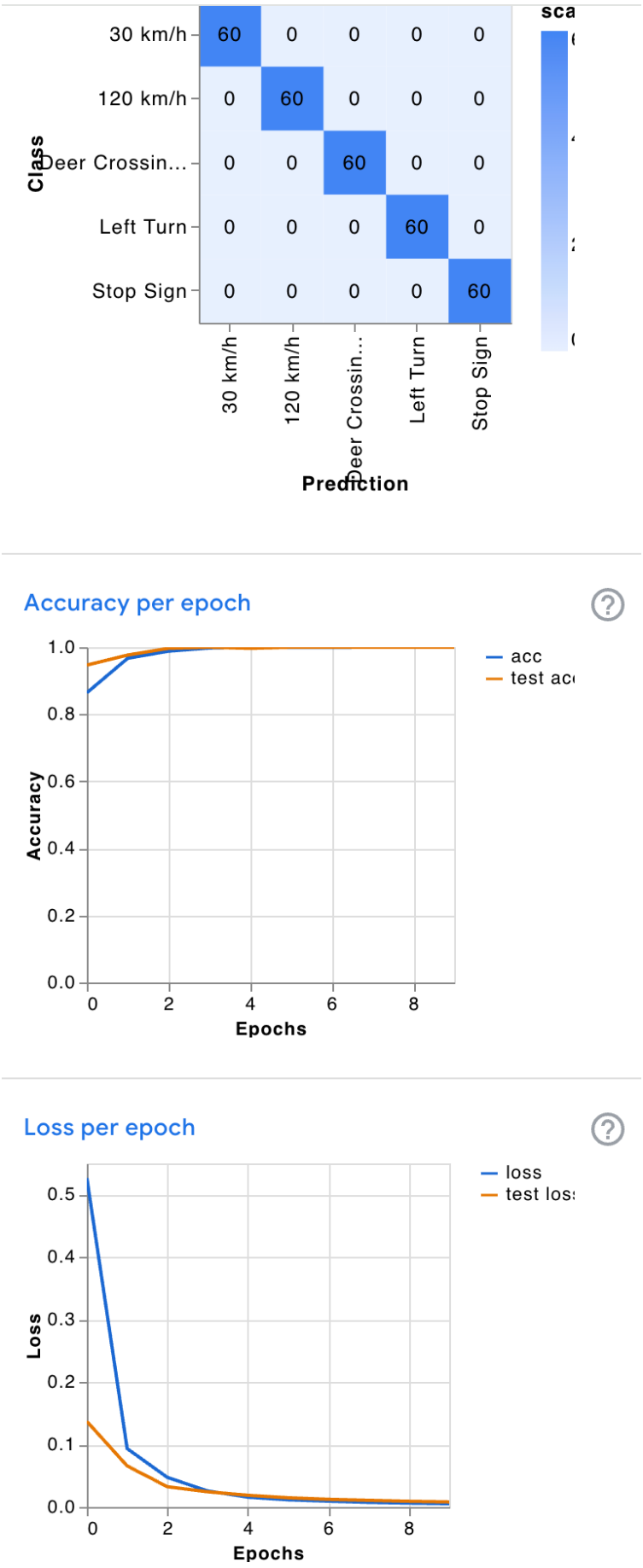


Figure 5 Combination